

Rugprint

Model/mode: GPT-5.5 Thinking

Project type: data visualization vignette / small nbdev package

One-line idea

Rugprint is a compact visualization for comparing how groups occupy multivariate measurement space.

It combines small scatterplots with marginal rug marks so that each panel shows both:

1. the pairwise relationship between two variables, and
2. the one-dimensional distribution of each variable within each group.

The first demo uses the Palmer Penguins dataset, but the same visualization pattern is intended for behavioral phenotyping, biological screens, and other “many metrics, few groups” datasets.

Motivation

Many biological and behavioral datasets are not well summarized by a single metric. A group may differ in one dimension, overlap in another, or occupy a different region of feature space only when two measurements are viewed together.

Standard scatterplots show pairwise structure, but they often hide the marginal distributions. Standard rugplots show distribution along one axis, but they are usually used as decoration rather than as a compact state-profile.

Rugprint treats the rug marks as part of the main comparison: each panel becomes a small fingerprint of how groups distribute across two measurements.

Example dataset

The demo uses Palmer Penguins because the variables are familiar and interpretable:

- bill_length_mm
- bill_depth_mm
- flipper_length_mm
- body_mass_g
- species
- sex
- island

For example, a Rugprint plot can compare penguin species across:

- bill length vs bill depth
- bill length vs flipper length
- body mass vs flipper length
- body mass vs bill depth

This makes the demo easy to understand without relying on unpublished or domain-specific data.

Core visualization

Each panel contains:

- a scatterplot of two measurements
- points colored by group
- x-axis rug marks showing the distribution of the first measurement
- y-axis rug marks showing the distribution of the second measurement
- optional ranking of feature pairs by group separation

The goal is not to replace statistical modeling. The goal is to provide a fast visual first pass over multivariate group structure.

Intended use cases

Rugprint is useful when you have:

- several quantitative features
- two or more groups or conditions
- interest in how groups differ across multiple measurements
- a need to avoid cherry-picking one metric too early

Possible applications:

- behavioral phenotyping
 - feeding assay metrics
 - pose-tracking outputs
 - morphology datasets
 - biological screening readouts
 - quality-control summaries
 - exploratory data analysis for small-to-medium feature tables
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Why not just use pairplot?

A pairplot is useful, but it can become visually dense. Rugprint is deliberately more curated:

- select a small number of meaningful feature pairs
- show each pair as a readable panel
- keep marginal distributions visible
- make the output suitable for a README, blog post, or paper supplement

Rugprint is closer to a compact visual fingerprint than an exhaustive exploratory matrix.

Minimal API

```
from rugprint.core import load_penguins, rugprint, rank_pair_separation

penguins = load_penguins()

fig = rugprint(
    penguins,
    pairs=[
        ("bill_length_mm", "bill_depth_mm"),
        ("bill_length_mm", "flipper_length_mm"),
        ("body_mass_g", "flipper_length_mm"),
        ("body_mass_g", "bill_depth_mm"),
    ],
    group="species",
)
```

Optional helper:

```
features = [
    "bill_length_mm",
    "bill_depth_mm",
    "flipper_length_mm",
    "body_mass_g",
]

rank_pair_separation(
    penguins,
    features=features,
    group="species",
)
```

nbdev structure

```
rugprint/  
  nbs/  
    index.ipynb  
    00_core.ipynb  
  rugprint/  
    core.py  
  README.md  
  pyproject.toml
```

`00_core.ipynb` exports the plotting functions.

`index.ipynb` becomes the project landing page and demo.

MVP checklist

- ☐ Create nbdev repo
 - ☐ Add `load_penguins()`
 - ☐ Add `rugprint()`
 - ☐ Add `rank_pair_separation()`
 - ☐ Generate one polished penguins demo figure
 - ☐ Write README from `index.ipynb`
 - ☐ Add screenshot to GitHub repo
 - ☐ Link from personal website under “Data visualization” or “Selected work”
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Portfolio framing

Rugprint is a small project, but it demonstrates a larger measurement philosophy:

Before reducing a complex state to one number, look across several measurements at once.

This connects naturally to estimation graphics, behavioral state inference, ethomics, and uncertainty-aware visualization.